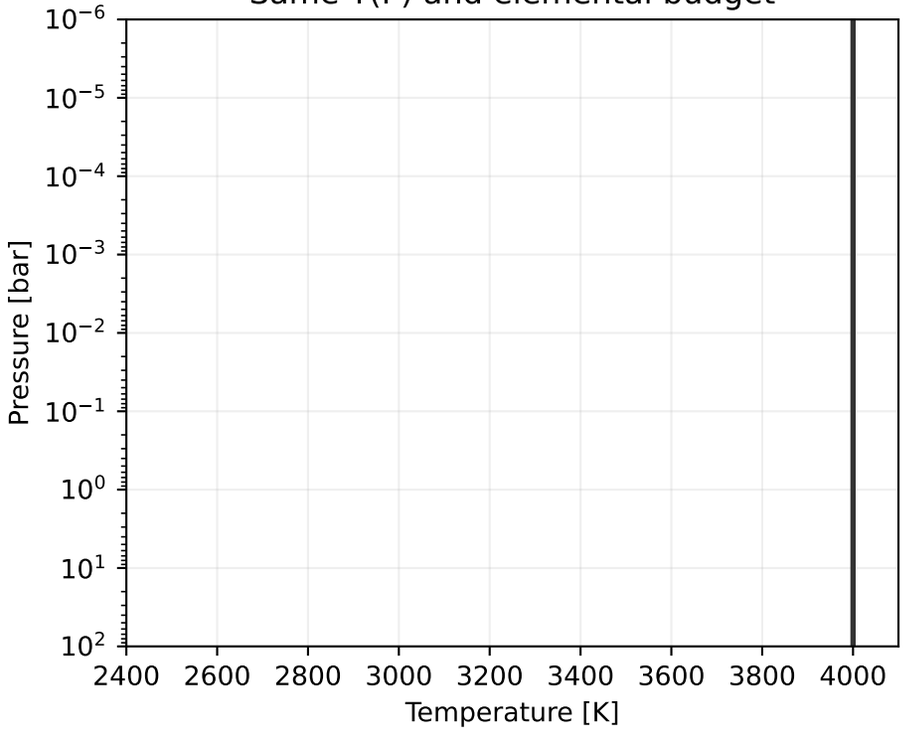
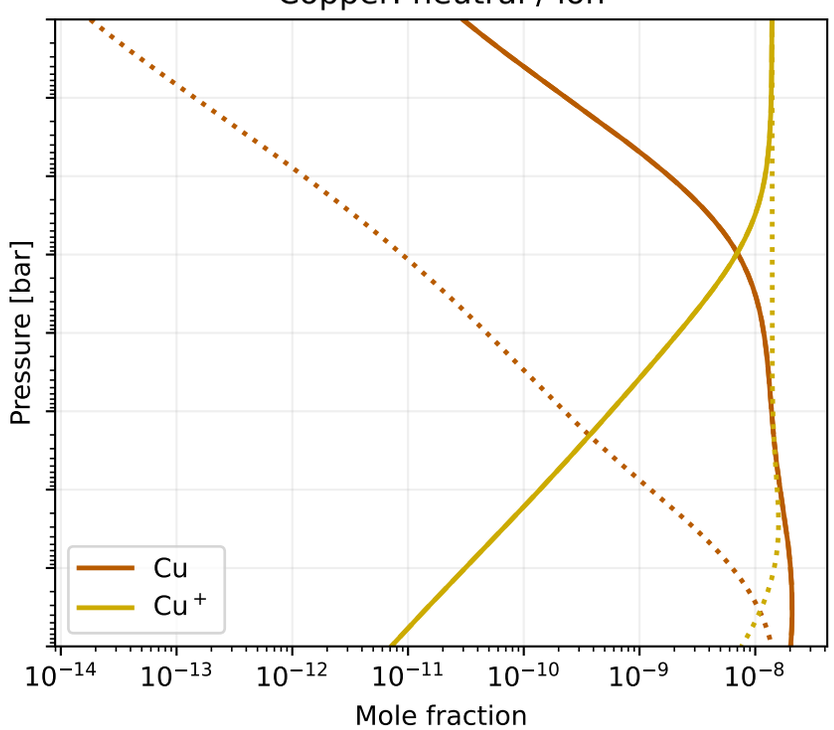
4000 K isothermal column | solar composition, same 537 species

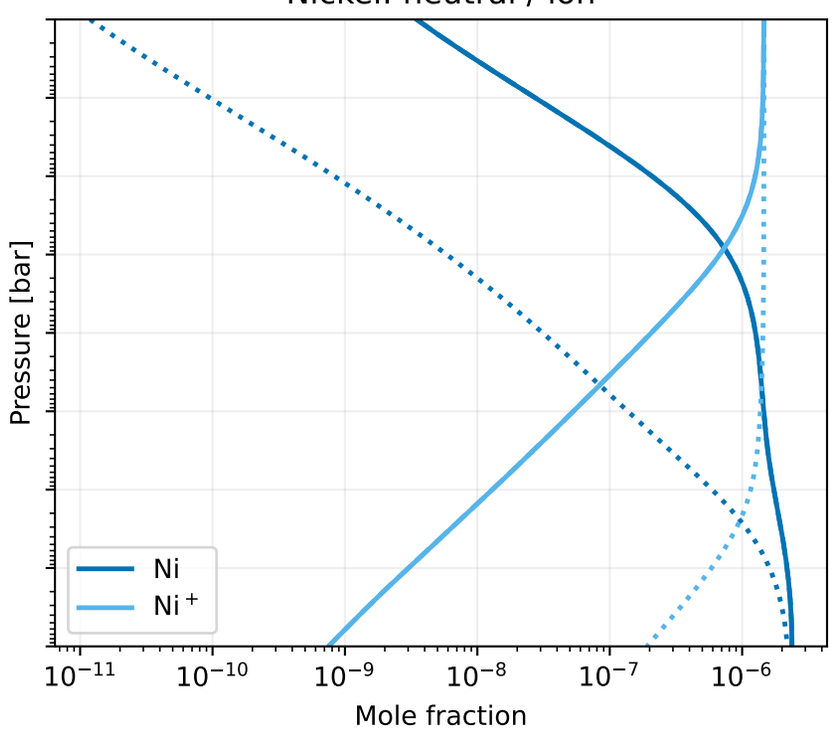
Same T(P) and elemental budget



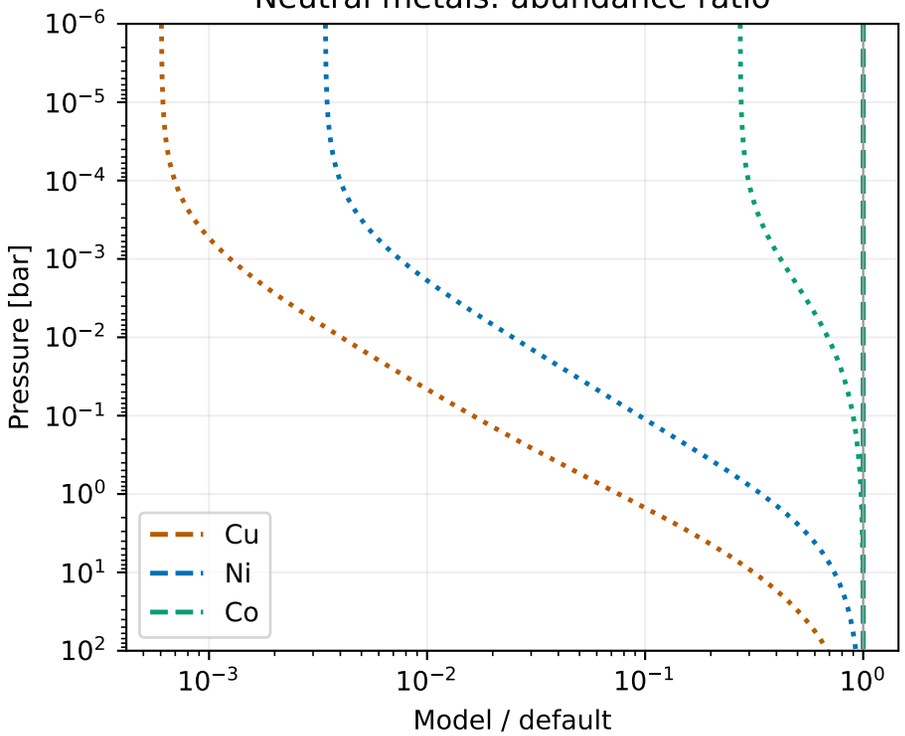
Copper: neutral / ion



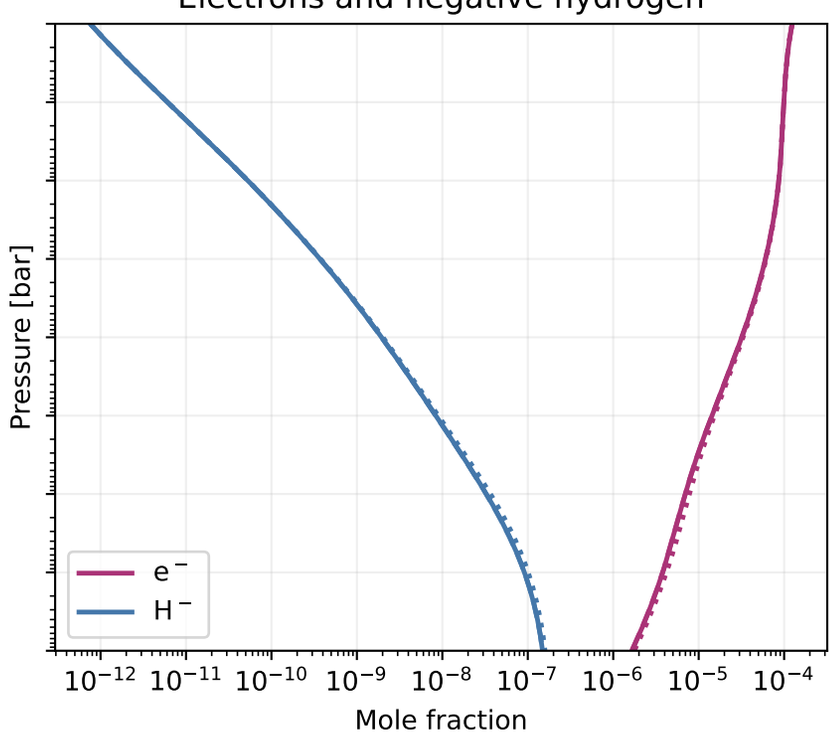
Nickel: neutral / ion



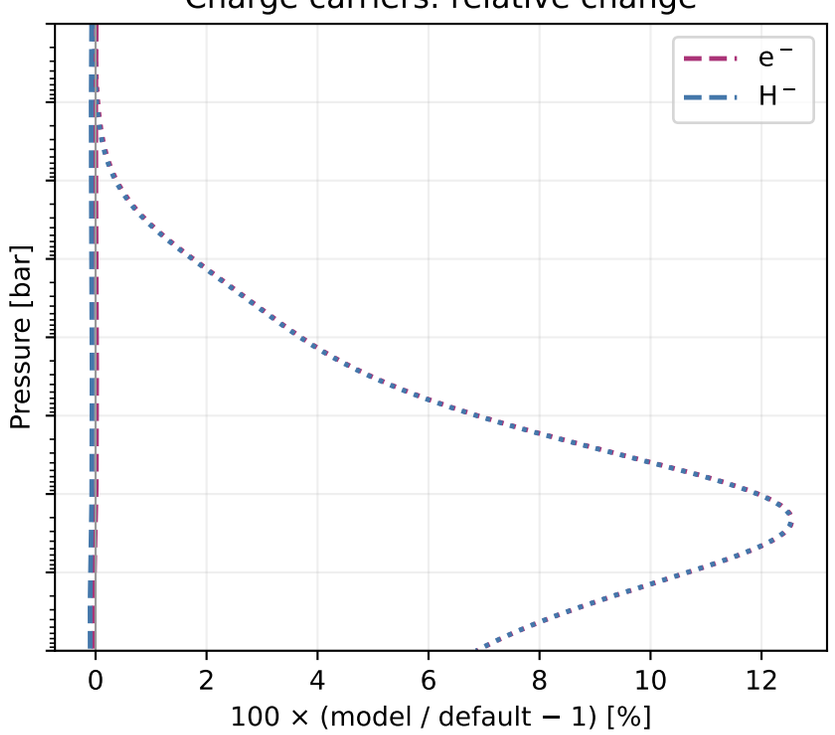
Neutral metals: abundance ratio



Electrons and negative hydrogen



Charge carriers: relative change



ExoGibbs default (curated) — New FastChem data (preliminary) ····· Original FastChem 4 data

Ratio panels: dashed = new FastChem / default; dotted = original FastChem / default.
ExoGibbs solver for every curve; prescribed ideal-gas columns, without condensation or photoionization.